USN | | | | | | | | | | | [22AS44]

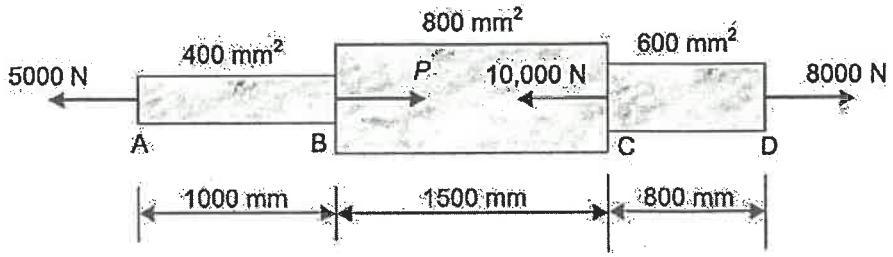
FOURTH SEMESTER B.Tech EXAMINATION, MAY - 2024

MECHANICS OF SOLIDS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART - A	**05x02=10**		
1.	**a.**	Define the poison's ratio.	1	1	**02**
	b.	Explain the Proof Stress with the help of the stress-strain curve.	2	1	**02**
	c.	Explain the different types of loads applying on the beams with the help of neat sketches	2	2	**02**
	d.	Obtain the Section Modulus for a rectangular section	3	3	**02**
	e.	Obtain the polar moment of inertia for solid and hollow circular shafts.	3	4	**02**
		PART – B	**06x05=30**		
2.	**a.**	Explain every point on the stress-strain relationship curve for mild steel with the help of neat sketches.	2	1	**05**
		OR			
	b.	The following observations were made during a tensile test on a mild steel specimen of 40 mm diameter and 200 mm long: Elongation with 40,000 N load (within the limit of proportionality) = 0.0304 mm, Yield load = 165,000 N, Maximum load = 245,000 N, Length of the specimen at fracture = 252 mm, Obtain the yield stress, the modulus of elasticity, the ultimate stress and the percentage elongation.	2	1	**05**
3.	**a.**	A steel bar ABCD of varying cross-section is subjected to the axial forces as shown in Fig. Obtain the value of *P* for equilibrium. If the modulus of elasticity $E = 2.1 \times 10^5$ N/mm^2, determine the elongation of the bar. 	2	1	**05**
		OR			
	b.	Obtain the Relation equation between E, K, and υ for a cube subjected to three equal stresses	2	1	**05**
4.	**a.**	Explain the relationship between load intensity, shear force, and bending moment with the help of neat sketches	2	2	**05**

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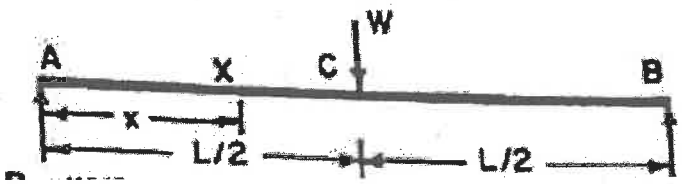
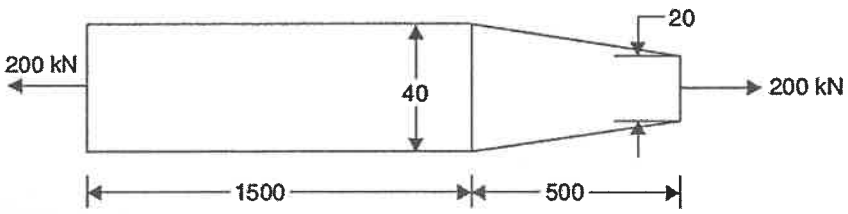
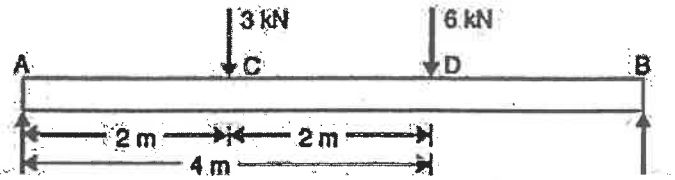
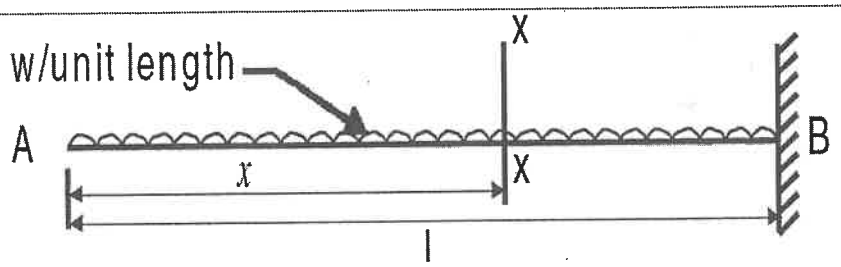
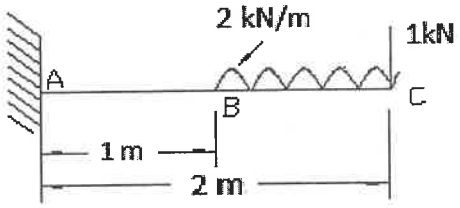
		OR			
	b.	Draw the shear force and bending moment diagram for the beam loaded as shown in fig. 	3	2	**05**
5.	**a.**	Obtain the equation M/I= σ/y = E/R from theory of simple bending to get the relation between various characteristics of a beam.	2	3	**05**
		OR			
	b.	Obtain the maximum deflection of simply supported beam subjected to a simple bending.	2	3	**05**
6.	**a.**	A bar subjected to a tensile force of 12 kN, if the cross section is 10 mm^2, solve for the Normal, tangential and resultant stress, intensity of plane making an angle of 35^0 with the plane of original stress.	3	5	**05**
		OR			
	b.	The principal stress at a point in the section of a boiler shell are 80 Mpa and 40 Mpa both tensile. solve for the Normal, Tangential and resultant stress intensities across a plane through the point inclined at 50^0 to the plane having the 80 Mpa stress.	3	5	**05**
7.	**a.**	Choose the ratio of the strength of a solid steel column to that of a hollow of the same cross-section areas. The internal diameter of the hollow column is 3 / 4 of the external diameter. The columns have the same length and are pinned at the ends.	3	6	**05**
		OR			
	b.	Using Euler's formula, solve for the critical stress of a series of struts having slenderness ration of 40, 80, 120, 160, and 200 under the following conditions: **i.** Both ends hinged, **ii.** Both ends fixed Take E = 2.05×10^5 N/MM^2	3	6	**05**
		PART – C	**06x10=60**		
8.	**a.**	Obtain the total extension of the bar with varying rectangular cross-sections subjected to tensile loads.	2	1	**10**
		OR			

	b.	A bar of 30 mm diameter is subjected to a pull of 60 kN. The measured extension on gauge length of 200 mm is 0.1 mm and change in diameter is 0.004 mm. Solve for the following: **i.** Young's modulus, **ii.** Poisson's ratio and **iii.** Bulk modulus	3	1	**10**
9.	**a.**	2.0 m long steel bar has a uniform diameter of 40 mm for a length of 1 m and in the next 0.5 m its diameter gradually reduces from 40 mm to 20 mm as shown in Fig. Solve for the elongation of this bar when subjected to an axial tensile load of 200 kN. Given E = 200 GN/m^2.  	3	1	**10**
		OR			
	b.	A hollow steel tube is to be used to carry an axial compressive load of 160 kN. The yield stress for steel is 250 N/mm2. A factor of safety of 1.75 is to be used in the design. The following three class of tubes of external diameter 101.6 mm are available. Class Thickness Light 3.65 mm Medium 4.05 mm Heavy 4.85 mm, Which section is sutable?	3	1	**10**
10.	**a.**	A simply supported beam of length 6 m carries point load of 3 kN and 6 kN at distances of 2 m and 4 m from the left end. Draw the shear force and bending moment diagrams for the beam.  	3	2	**10**
		OR			
	b.	Draw the shear force and bending moment diagrams for the beam shown in fig.	3	2	**10**

					
11.	**a.**	Obtain the vertical deflection at free end C of the cantilever beam ABC shown in figure. Consider moment of inertia is 'I' and young's modulus is 'E'. 	2	3	**10**
		OR			
	b.	Obtain the maximum slope and deflection of simply supported beam length 'l', intensity of load 'W' applied at the center of the beam by double integration method	2	3	**10**
12.	**a.**	Obtain the maximum deflection of simply supported beam is not subjected to pure bending couple and the radius of curvature will not be constant at the all the points.	2	3	**10**
		OR			
	b.	Examine the dimensions of a solid shaft according to strength criteria and stiffness criteria to transmit the power 150 kW at 300 rpm. If the allowable shear stress 60 N/mm^2 and allowable twist is 1^0 in the length of 3.5 mts and modulus of rigidity $G = 82 \times 10^3$ N/mm^2.	4	4	**10**
13.	**a.**	Obtain the equation for the critical load of a column with one end fixed and the other hinged.	2	6	**10**
		OR			
	b.	A cylindrical shell is 3m long, 1-meter internal diameter and 15mm metal thickness. Solve for the maximum intensity of shell stress-induced and also the changes in the dimensions of the shell if it is subjected to an internal pressure of 1.5 N/mm^2. Take $E = 0.204 \times 10^6$ mm^2 and $1/m = 0.3$	3	6	**10**